Talyor Series expansion: Exercise 4

where: $x = x_0$ use:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + O((x-x_0)^3)$$

1. $f(x) = \frac{x}{x+1}$

$f'(x) = \frac{1}{(x+1)^2}$

$f''(x) = -\frac{2}{(x+1)^2}$

Derivatives, deriving the equation

$f(x_0) = \frac{x_0}{x_0+1}$

$f'(x_0) = \frac{1}{(x_0+1)^2}$

$f''(x_0) = -\frac{2}{(x_0+1)^3}$

Evaludte $x = x_0$

$$f(x) \approx \frac{x}{x+1} = \frac{x_0}{x_0+1} + \frac{x-x_0}{(x_0+1)^2} - \frac{(x-x_0)^2}{(x_0+1)^2} + O((x-x_0)^3)$$

2. $f(x) = \dfrac{2x^2}{x^4}$

Simplify The equation: $f(x) = 2x^{-2}$

$= \dfrac{2}{x^2}$ → Using the simplified equation to derive

$\therefore f(x) = \dfrac{2}{x^2}$

$f'(x) = -4x^{-3}$

$f''(x) = 12x^{-2}$

derivatives of the equation

$f(x_0) = \dfrac{2}{x_0^2}$

$f'(x_0) = -\dfrac{4}{x_0^3}$

$f''(x_0) = \dfrac{12}{x_0^4}$

Evaluate $x = x_0$

$$f(x_0) \approx \frac{2x^2}{x^4} = \frac{2}{x_0^2} - \frac{4}{x_0^2}(x - x_0) + \frac{6}{x_0^4}(x - x_0)^2 + O((x - x_0)^3)$$

3. $f(x) = \ln(x^2)$

Using : $\ln(x^2) = 2\ln x \quad (x > 0)$

$$\left.\begin{aligned} f'(x) &= \frac{2}{x} \\ f''(x) &= -\frac{2}{x^2} \end{aligned}\right\} \text{derivatives of the equation}$$

$$\left.\begin{aligned} f(x_0) &= \ln(x_0^2) \\ f'(x_0) &= \frac{2}{x_0} \\ f''(x_0) &= -\frac{2}{x_0^2} \end{aligned}\right\} \text{evaluating the equation where } x = x_0$$

$$f(x) \approx \ln(x^2) = \ln(x_0^2) + \frac{2}{x_0}(x - x_0) - \frac{1}{x_0^2}(x - x_0)^2 + O((x - x_0)^3)$$